

Shared evidence | Signals and conditions

Hormone messages: the control room

Name _____ Date _____

Original conceptual cases; no personal health or medical decision is involved.

Endocrine structure	Approximate position
Pituitary	Base of brain
Thyroid	Neck
Pancreas	Upper abdomen
Adrenal glands	Above the kidneys
Ovaries	Pelvis
Testes	Scrotum

Cell model A hormone is present near both cells A and B. A has a matching receptor; B lacks that receptor.	Control case An internal variable rises above its usual suitable range. In this model, the control response can reduce that departure.
Conditions case A cell’s enzyme-controlled reactions depend on suitable temperature. Water balance also affects animal-cell volume.	

W1. Trace a message. W2. Predict a responding cell. W3. Explain a corrective response.

Supported route | Label the mechanism

Word bank: endocrine gland; blood; target; receptor; response; suitable range.

S1. Put gland, blood and target cell in order. Say what happens at each step.

S2. Choose A or B from the cell model and explain the recognition difference.

S3. An internal variable rises. Describe the direction of a response that counteracts this rise.

S4. Give one reason suitable internal conditions matter. Locate the pancreas and thyroid.

Standard route | Explain the control room

Use the shared evidence and accurate biological terms.

M1. Explain how a hormone reaches a target organ, distinguishing release, transport and response.

M2. Explain why cell A can respond when cell B does not, although both are exposed.

M3. Explain homeostasis and negative feedback without claiming perfect constancy or that all responses use hormones.

M4. Name the positions of three endocrine glands and explain one reason internal temperature or water balance matters.

Stretch route | Test the model

Optional deeper reasoning; detailed Higher-tier hormone cascades are not required.

T1. A cell has a matching receptor but its response machinery is damaged. Is receptor presence alone enough to guarantee a normal response? Explain.

T2. A model response arrives late and the variable briefly passes its target. Does this prove there is no control? Explain one model limitation.

T3. Compare a broadly circulating chemical message with a directed nerve signal at a conceptual level. Avoid claiming all hormone responses are always slow.

T4. Explain why a general feedback diagram cannot predict a person's exact hormone level or diagnose illness.

Exit and reflection | Route, recognition, regulation

Use short explanations or labelled arrows.

E1. How do endocrine hormones travel to tissues?

E2. Why may a cell fail to respond to a particular hormone?

E3. What does homeostasis regulate, and why?

L1. What did your audience ask? Record one improved or defended mechanism.

Next useful step

- A. Locate an endocrine gland.
- B. Trace transport in blood.
- C. Explain receptor specificity.
- D. Describe regulation within limits.